

Cuarto Exam Parcial

3) $m = 18 \text{ Kg}$ $\theta = 20^\circ$ $d = 20 \text{ m}$
 $\mu = 0.5$

$$T_x = T \cos 20^\circ$$

$$\sum F_x = 0$$

$$T_x - F_r = 0$$

$$T_x = F_r$$

$$F_r = \mu N$$

$$T \cos 20^\circ = \mu N$$

$$\sum F_y = 0$$

$$N + T_y - P = 0$$

$$N = mg - T \sin 20^\circ$$

$$T \cos 20^\circ = \mu mg - \mu T \sin 20^\circ$$

$$T = \frac{\mu mg}{\cos 20^\circ + \mu \sin 20^\circ} = \frac{0.500 \cdot 18 \cdot 9.8}{\cos 20^\circ + 0.5 \sin 20^\circ} = 79.97$$

$$W = T_x d \cos 20^\circ$$

$$W = T \cos 20^\circ d$$

$$W = 79.97 \cos 20^\circ \cdot 20 = 1492.42 \text{ joules}$$

$$4) m = 20g = 0.02Kg \quad g = 9.8m/s^2$$

$$y(A) = -0.120m$$

$$y(B) = 0$$

$$y(C) = 20m$$

$$K(C) + U_g(C) + U_s(C) = K(A) + U_g(A) + U_s(A)$$

$$0 + mgy(C) + 0 = 0 + mgy(A) + \frac{1}{2} K y(A)^2$$

$$K = \frac{2mg[y(C) - y(A)]}{y(A)^2} = \frac{2(0.02)(9.8)[20 - (-0.120)]}{(-0.0120)^2} =$$

$$547.77 N/m$$

5)

$$P = 120W$$

$$x = 20m$$

$$W = 1492.42$$

$$\text{Trabajo} = (F \cos \theta) x$$

$$1492.42 = F \cos(20) 20$$

$$F = 79.41$$

$$P = \frac{F_x}{t} = P = FV$$

$$120 = 79.41 V$$

$$V = 1.51$$